

EXECUTION CONTEXT

Interviewquestion-158



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What is Execution Context?

Execution context refers to the environment in which JavaScript code is executed

Global execution context

A large yellow rectangle occupies the lower half of the slide, representing the global execution context. It is positioned below a white header bar that contains the text "Global execution context".

Understanding Execution Context: The Two Phases of JavaScript Execution

In JavaScript, the concept of execution context is fundamental to understanding how the language operates.

Every time a script runs, it goes through two main phases:

- The Memory Creation Phase and
- The Code Execution Phase.

Global execution context

Memory Creation Phase

code Execution Phase

Memory Creation Phase:

During this phase, the JavaScript engine allocates memory for variables and functions before executing any code. This is often referred to as "hoisting."

Here's what happens:

- Variables are declared and initialized to undefined.**
- Functions are stored in memory in their entirety.**

Example

```
var name = "Alice";  
function greet(person) {  
    var greeting = "Hello, " + person + "!";  
    return greeting;  
}  
  
var greetingForAlice = greet(name);
```

memory execution phase

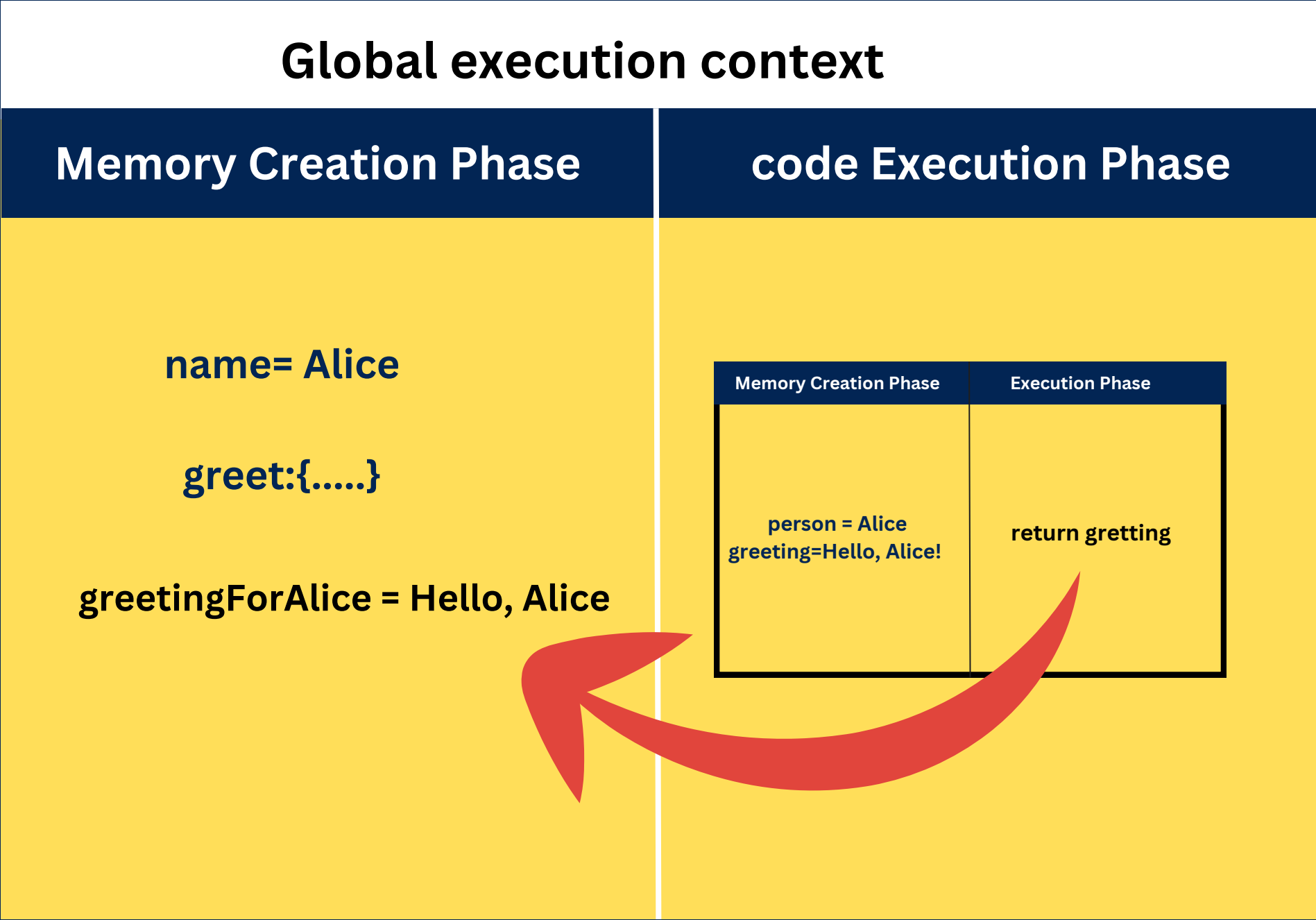


Memory creation phase is executed for above example

Global execution context	
Memory Creation Phase	code Execution Phase
name= Alice greet :{.....} greetingForAlice = undefined	

Execution Phase:

Once memory is allocated, the code execution phase begins. The JavaScript engine **executes the code line by line, assigning values to variables and invoking functions as defined.**



To clearly understand the above, each function call creates a new execution context within the existing execution context. It follows the same memory allocation phase and code execution phase.

Explaining the function call execution phase:

```
function greet(person) {  
    var greeting = "Hello, " + person + "!";  
    return greeting;  
}
```

Memory Phase

Memory Creation Phase

code Execution Phase

person = undefined

greeting = undefined

Code ExecutionPhase

Memory Creation Phase	code Execution Phase
person = Alice greeting = Hello alice	return Hello alice

The program completes when control returns to the global execution context, and all code has been executed. The execution happens based on the code structure, with deeply nested executions passing control back to their parent contexts, eventually returning to the global execution context. Once the global execution context completes, the entire code execution is finished.



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